

You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

For problems 1 through 8, find all real solutions to the following equations. Write all answer as exact values

1. $\frac{3x}{2} - 1 = 2 - \frac{7x}{4}$

5. $\frac{z+2}{z-3} = \frac{5}{-3}$

2. $\frac{1}{2}|x-9| - 7 = 9$

6. $\sqrt{x+3} = x+1$

3. $C = Ax + By$ for y .

7. $2e^{x-1} = 10$

4. $x^2 - 11 = 0$

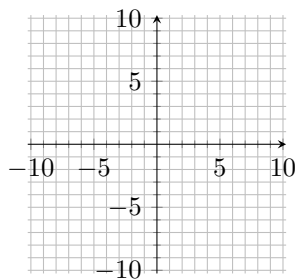
8. $\log_3(4) = 3 - \log_3(x)$

Solve the word problems. Approximate your answer to three decimal places.

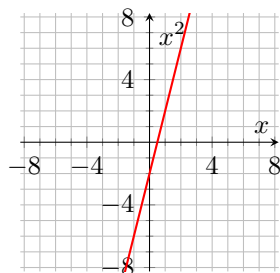
9. Jeff knows that Holly paid \$40,230, including sales tax for a new Buick La-Sabre. If the sales tax is 10%, then what is the cost of the car *before* the tax?

10. How long will it take \$100 to grow to \$350 if it is invested at 8% compounded continuously?

11. Find the center and the radius of the circle. Then graph the circle $(x - 2)^2 + (y + 4)^2 = 16$.



12. Write an equation for the line shown in the graph.



For problems 13-14, perform the indicated operations and write your answers in the form $a + bi$.

13. $(2 - i)(4 + 8i)$

14. Simplify the following:

- a. i^4
- b. i^{25}

15. Let $h = \{(1, 2), (2, 2), (3, 2), (4, 2)\}$ be a relation assigning x coordinates with y coordinates.

- Domain:
- Range:
- Is h a function of x ?

Let $f(x) = x + 1$ and $g(x) = x^2 - x$. Find and simplify the expression.

16. $g(-2)$

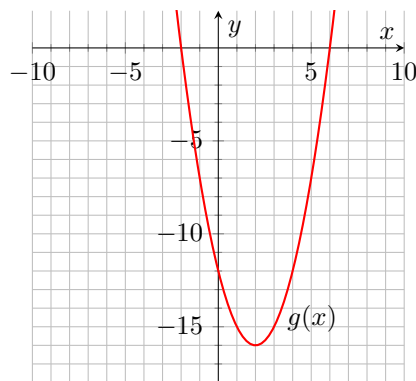
17. $(g \circ f)(x)$

Use the difference quotient, $\frac{f(x+h) - f(x)}{h}$ for $h \neq 0$, to complete and simplify the following problem.

18. $f(x) = 2x + 1$.

19. [2pts each] Use the graph of the quadratic function, to answer the following:

- What is the vertex?
- What is the axis of symmetry?
- What is the range?
- If any exist, find the x -int(s).
- Find the y -int.
- What are the zeros of $f(x)$?
- Find the simplest polynomial with real coefficients that has the roots from above (you do not have to expand).



20. Consider the polynomials $P(x) = 2x^2 - 3x + 6$ and $(x - 5)$.

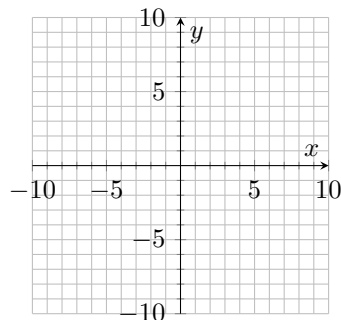
- a. Use division (synthetic or traditional) to find the quotient and remainder when $P(x)$ is divided by $x - 5$.

b. Find $\frac{P(x)}{x - 5}$

- c. [2 pts] Is $x = 5$ a root/zero of $P(x)$?

21. [2 pts each] Use $f(x) = \frac{2x + 1}{x + 3}$

- a. Find the domain of $f(x)$.
- b. Find any vertical asymptotes.
- c. Find the horizontal asymptote.



- d. Sketch a graph of the function

22. [2pts each] Find the following:

a. $x^5 - 10000$ as $x \rightarrow \infty$

b. $\lim_{x \rightarrow \infty} \frac{1}{x}$

c. $\lim_{x \rightarrow -3^+} \frac{2}{x + 3}$

d. The domain of $\frac{3x^{13} - 12x + \sqrt{5}}{(x + 3)(x - 1)}$

23. Evaluate the following:

a. $\log_6 (6^2)$

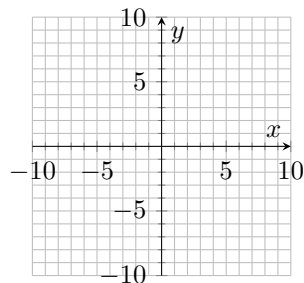
b. $\frac{1}{3}e^{\ln(3)}$

24. Let $g(x) = \log_2 (x - 2)$.

a. Graph $g(x)$.

b. Find the domain and range of $g(x)$.

c. Graph the inverse of $g(x), g^{-1}(x)$.



25. Use the properties of logarithms to expand the expression as much as possible.

$$\log_{11} (11y^2)$$

26. Use the properties of logarithms to rewrite the expression as a single logarithm.

$$\ln(4) + 2 \ln(3)$$

27. Find and graph the equation of the parabola with focus $(-1, 6)$ and directrix $y = 2$.

